

PUNJAB SILAGE FOR FEED STORAGE

Sector: Agriculture

Hazard addressed: Droughts

Targeted assets: Buffalo, Cattle, Goat, Sheep



Description: This measure involves production and storage of silage as preserved livestock feed to ensure fodder availability during droughts, dry seasons, and climate-induced fodder shortages. The intervention supports long-term feed storage, improves livestock nutrition, stabilizes milk and meat production, and enhances resilience of livestock systems against climate variability.

Priority Districts: Bahawalnagar, Bahawalpur, Dera Ghazi Khan, Khanewal, Lodhran, Multan, Rahim Yar Khan, Rajanpur, Muzaffargarh, Bhakkar, Mianwali, Khushab, Chakwal, Toba Tek Singh, Vehari.

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Animal per year.

Scale assumptions: Applicable in livestock-dependent and drought-prone districts vulnerable to seasonal fodder shortages and declining grazing resources.

Key design parameters: Silage preparation using maize, sorghum, or fodder crops; airtight storage systems; moisture control; feed quality management; and farmer training on silage preparation and utilization.

Time horizon: Done in three batches (2034 – 2035, 2039 – 2040, 2044 – 2045).

COST ASSUMPTIONS

Unit cost:
PKR 44,500 per animal/year (~USD 159 per animal/year).

Capital Expenditure (CAPEX):
Average USD 12 million in the first year. Includes silage pits/bags, storage infrastructure, chopping equipment, and initial setup costs.

Operational Expenditure (OPEX):
Average USD 108 million in the second year. Includes fodder crop production/purchase, silage preparation, labor, maintenance, and feed management.

IMPACT

Risk reduction level: High.

Time to effectiveness: Immediate to short-term following silage preparation and storage.

Expected impact: Improved livestock feed availability during drought periods, reduced fodder shortages, improved livestock productivity, and enhanced resilience of livestock systems against climate-induced stress.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Farmer training, access to fodder crops, silage preparation equipment, extension services, and financing support.

Required institutions: Livestock Department Punjab, Agriculture Department Punjab, Dairy Development Agencies, Agricultural Extension Services, and development partners.

Barriers: Limited technical awareness, high recurring fodder costs, inadequate storage facilities, and weak extension support systems.

CO-BENEFITS & TRADE-OFFS

Environmental: Reduced pressure on degraded grazing lands and improved fodder resource management.

Social: Improved livelihood resilience and food security for livestock-dependent households.

Economic: Reduced livestock productivity losses during droughts and improved income stability for livestock farmers.

Potential risks: Poor silage preparation practices, spoilage due to inadequate storage, and dependence on availability of fodder crops.

BENEFICIARIES

Primary beneficiaries: Livestock farmers, dairy producers, pastoral communities, and rural households dependent on livestock production.

Distributional aspects: Particularly beneficial for drought-prone livestock-dependent communities vulnerable to fodder shortages.